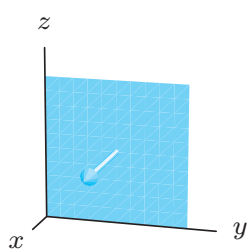
Suggested problems: 1-9 odd, 13-25 odd, 31-47 odd

Problem 1: Suppose $\vec{F}(x, y, z) = \langle 4, 0, 5 \rangle$ and S is the square of side length 3 perpendicular to the z-axis, centered at $(0, 0, -2)$, and oriented toward the origin. Compute the flux of $\vec{F}$ through S.

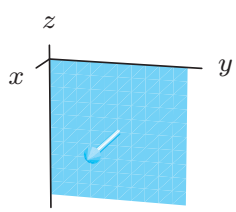
Problem 2: Suppose $\vec{F}(x, y, z) = \langle 2, 0, 3 \rangle$ and S is the disk of radius 4 perpendicular to the x-axis, centered at $(5, 0, 0)$, and oriented toward the origin. Compute the flux of $\vec{F}$ through S.

Problem 3: Let $\vec{F}(x, y, z) = \langle z, 0, 0 \rangle$. For each of the oriented surfaces below, decide whether the flux of $\vec{F}$ is positive, negative, or zero.

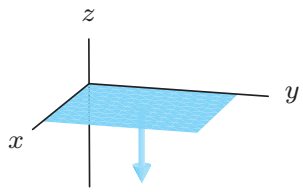
(a)



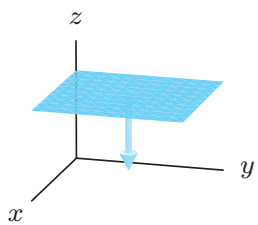
(b)



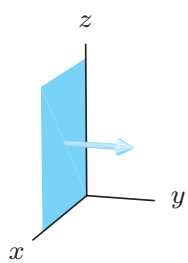
(c)



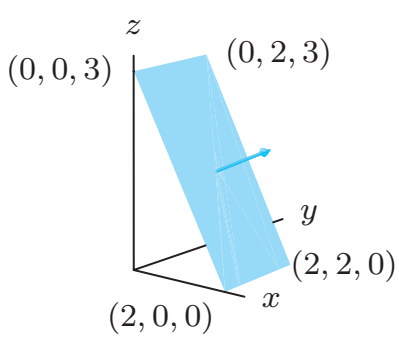
(d)



(e)



Problem 4: Suppose $\vec{F}(x, y, z) = \langle 1, 2, -3 \rangle$ and that S is the oriented surface below. Compute the flux of F through S.



Problem 5: Let $\vec{F}(x, y, z) = \langle 1, -2, z \rangle$, and suppose S is the part of the graph of $f(x, y) = xy$ lying above the region D which is $0 \leq x \leq 10$ and $0 \leq y \leq 10$. Compute $\int_S \vec{F} \cdot d\vec{A}$.

Problem 6: Let $\vec{F}(x, y, z) = \langle x, 0, z \rangle$, and suppose S is the part of the graph of $f(x, y) = x + y + 2$ lying above the region D which is the triangle with vertices $(0, 0, 0)$, $(1, 0, 0)$, and $(0, 1, 0)$. Compute $\int_S \vec{F} \cdot d\vec{A}$.